

Automatic Street Light Using LDR and Arduino

Objective

The main objective of this project is to create and simulate an automatic street light system using an Arduino Uno and a Light Dependent Resistor (LDR). The system is designed to switch the street light ON automatically when it becomes dark and turn it OFF when sufficient light is available. This helps in reducing energy consumption and eliminates the need for manual control.

Components Used

- Arduino Uno R3
- Light Dependent Resistor (LDR)
- LED
- 220Ω Resistor
- 10kΩ Resistor
- Connecting Wires
- Tinkercad Simulation Platform

Circuit Description

The circuit is built using an Arduino Uno, an LDR, an LED, and two resistors. The LDR and the 10kΩ resistor form a voltage divider circuit, which is connected to the Arduino's analog pin A0. The LED is connected to digital pin 13 through a 220Ω resistor to limit the current and protect the LED. The Arduino continuously monitors the light intensity through the LDR and controls the LED based on the detected light level.

Working Principle

An LDR is a sensor whose resistance changes depending on the amount of light falling on it. When the surrounding environment is bright, the resistance of the LDR decreases, producing a higher voltage reading at the Arduino's analog input. In this condition, the Arduino keeps the LED turned OFF. When the environment becomes dark, the resistance of the LDR increases, causing the analog reading to drop. Once the reading falls below a predefined threshold value, the Arduino automatically switches the LED ON. This process mimics the functioning of real-world automatic street lighting systems.

Applications

- Automatic street lighting
- Energy-saving outdoor lighting systems
- Smart city projects
- Garden and pathway illumination
- Security and safety lighting

Result

The Automatic Street Light system was successfully designed and tested using the Tinkercad simulation platform. The LED turned ON during low-light conditions and turned OFF when the light intensity increased. The project successfully demonstrated how sensors and Arduino-based automation can be used to develop simple, efficient, and energy-saving lighting solutions.